FlashMo: Geometric Interpolants and Frequency-Aware Sparsity for Scalable Efficient Motion Generation

Zeyu Zhang[1] Yiran Wang[1] Danning Li[2] Dong Gong[3] Ian Reid[2] Richard Hartley[1]

[1]ANU [2]MBZUAI [3]UNSW

Abstract

Diffusion models have recently advanced 3D human motion generation by producing smoother and more realistic sequences from natural language. However, existing approaches face two major challenges: high computational cost during training and inference, and limited scalability due to reliance on U-Net inductive bias. To address these challenges, we propose **FlashMo**, a frequency-aware sparse motion diffusion model that prunes low-frequency tokens to enhance efficiency without custom kernel design. We further introduce *MotionSiT*, a scalable diffusion transformer based on a joint-temporal factorized interpolant with Lie group geodesics over $\mathrm{SO}(3)$ manifolds, enabling principled generation of joint rotations. Extensive experiments on the large-scale MotionHub V2 dataset and standard benchmarks including HumanML3D and KIT-ML demonstrate that our method significantly outperforms previous approaches in motion quality, efficiency, and scalability. Compared to the state-of-the-art 1-step distillation baseline, FlashMo reduces **12.9%** inference time and FID by **34.1%**.

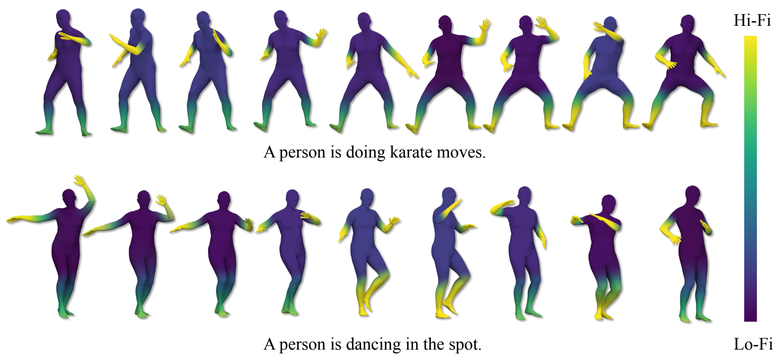


Figure 1. **Frequency of human motion.** The diagram illustrates that dynamic motions exhibit higher frequencies, while static motions correspond to lower frequencies. This observation provides key insights for the frequency-aware sparsification design of our FlashMo.

Challenges

- **Efficiency**: Motion diffusion models are computationally expensive and slow during both training and inference. Existing acceleration methods often rely on teacher models or complex mechanisms, leading to overengineering and limited efficiency.
- **Scalability**: Most motion diffusion models still depend on U-Net architectures and small datasets, limiting their scalability, while transformer-based designs show greater potential for large-scale generation.

Observation

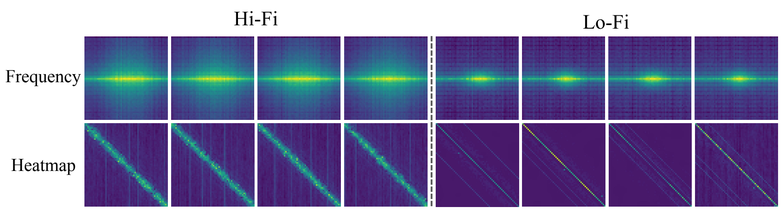


Figure 2. **Frequency magnitude vs. attention heatmap of attention heads.** The frequency magnitude is computed with the Fast Fourier Transform (FFT) and averaged over 100 latent motion features. Brighter colors indicate higher magnitudes. Pixels closer to the center represent lower frequencies. Both maps are visualized from the last layer of MotionSiT.

Frequency-Aware Sparsification

We design MotionSiT with a frequency-aware sparsification mechanism that dynamically prunes tokens corresponding to low-frequency motion to enhance efficiency, while preserving high-frequency motion at the attention head level. This structured sparsification enables seamless adaptation to hardware-efficient attention, such as FlashAttention, without any customized kernels.

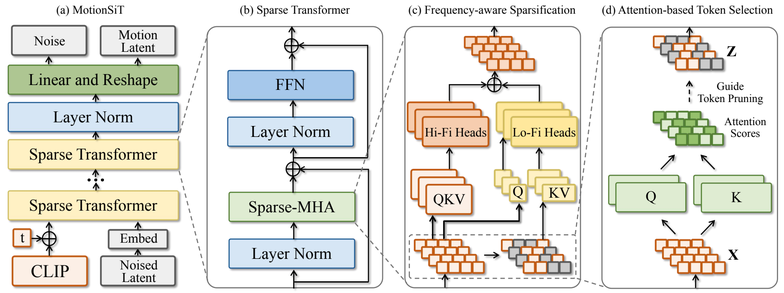


Figure 3. **FlashMo architecture.** FlashMo leverages frequency-aware sparsification and a MotionSiT backbone to efficiently and scalably generate motion from noised latent inputs.

Geometric Interpolants

Temporal-Spatial Factorized Interpolant applies independent noise schedules along the temporal and spatial dimensions. Specifically, we define:

$$x_t = \alpha_T(t) \odot \alpha_S(t) \odot x^* + \sigma(t)\epsilon$$

where $\alpha_T(t) \in \mathbb{R}^T$ and $\alpha_S(t) \in \mathbb{R}^S$ are scalar schedules for temporal and spatial dimensions, respectively. The element-wise product $\odot$ is broadcasted to modulate x^* at each timestep and spatial location individually.

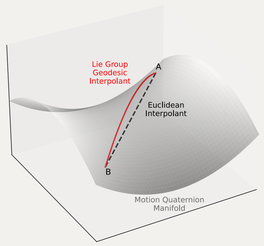


Figure 4. Euclidean interpolant vs. Lie group geodesics interpolant.

Lie Group Geodesics Interpolant respects the underlying manifold structure through Lie group geodesics. Given a target motion sample $x^* \in \mathcal{G}$, where $\mathcal{G}$ denotes a Lie group, we first map x^* to its tangent space $\mathfrak{g}$ using the logarithmic map $\log : \mathcal{G} \to \mathfrak{g}$. We then interpolate within the tangent space before mapping back to the manifold with the exponential map:

$$x_t = \mathrm{Exp}(\alpha(t) \cdot \log(x^*) + \sigma(t) \cdot \epsilon)$$

where ϵ is Gaussian noise in the Lie algebra $\mathfrak{g}$. This approach ensures that the interpolated sample x_t always lies on the manifold and that noise is injected in a way that respects the local geometry of the motion space.

Geometric Factorized Interpolant combines the two approaches above. Specifically, for motion representation lies on a manifold such as $\mathrm{SO}(3)$, we perform geodesic interpolant on both temporal and spatial dimension:

$$x_t^{\tau,s} = \mathrm{Exp}(\alpha_T(t)_\tau \cdot \alpha_S(t)_s \cdot \log(x^{*\tau,s}) + \sigma(t) \cdot \epsilon^{\tau,s})$$

where $x^{*\tau,s}$ denotes the motion state at time τ and spatial location s, and $\epsilon^{\tau,s}$ is Gaussian noise in the corresponding tangent space. This formulation combines the benefits of temporal-spatial control while preserves the geometric consistency of joint rotations during interpolation.

Scalability and Efficiency

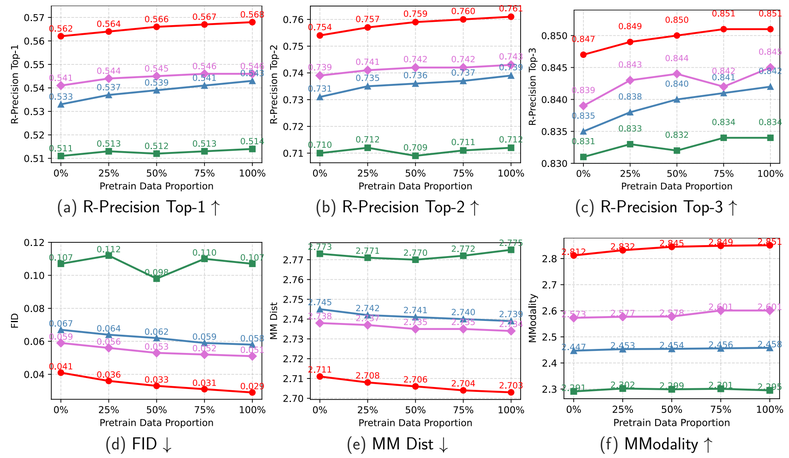


Figure 5. **Scaling trend.** The figure demonstrates the scaling trends of different denoiser designs (U-Net, DiT, SiT, and **MotionSiT**) with varying proportions of pretraining data. The results show that our MotionSiT exhibits superior scalability and outperforms other methods.

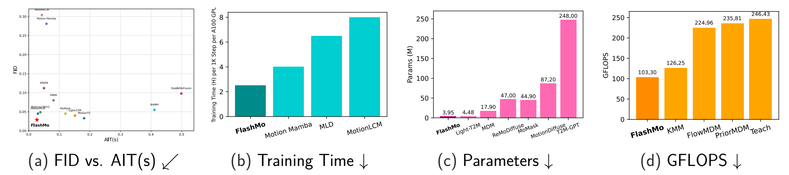


Figure 6. **Efficiency comparison.** The figure demonstrates that FlashMo achieves the lowest inference time, training time, model size, and FLOPs while maintaining superior performance compared to other methods.

Qualitative Results

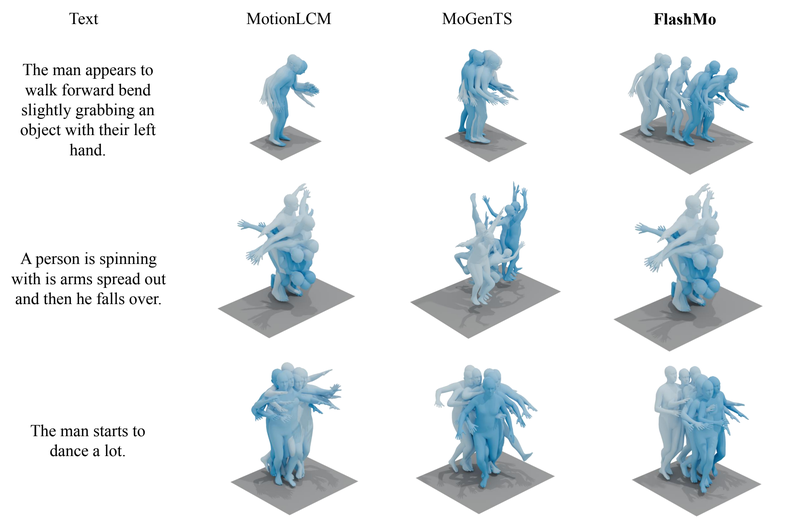


Figure 7. **Qualitative evaluation on HumanML3D test set.** We qualitatively compared the visualizations generated by our method with those produced by MotionLCM and MoGenTS.